MATH 415 - FALL 2020 - EXTRA PRACTICE PROBLEMS - WEEK 9

1. *For each of the following matrices, find the characteristic polynomial, eigenvalues and corresponding eigenvectors (i.e., for each eigenvalue give a maximal set of linearly independent eigenvectors which correspond to that eigenvalue):*

(1) $\begin{bmatrix} 3 & -1 \\ 1 & 5 \end{bmatrix}$

(2) $\begin{bmatrix} 4 & 0 & -2 \\ 2 & 5 & 4 \\ 0 & 0 & 5 \end{bmatrix}$

(3) $\begin{bmatrix} 4 & 0 & 0 & 0 \\ 0 & 4 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 1 & 0 & 0 & 2 \end{bmatrix}$

(4) $\begin{bmatrix} 4 & 3 \\ -3 & 4 \end{bmatrix}$

(1) We first find the characteristic polynomial:

$$p(\lambda) = \begin{vmatrix} 3-\lambda & -1 \\ 1 & 5-\lambda \end{vmatrix} = (3-\lambda)(5-\lambda) + 1 = \lambda - 8\lambda + 16 = (\lambda - 4)^2$$

We conclude that $\lambda = 4$ is an eigenvalue with multiplicity 2. We find the corresponding eigenvectors by computing a basis of the nullspace of $A - 4I$:

$$\begin{bmatrix} -1 & -1 \\ 1 & 1 \end{bmatrix} \xrightarrow{R_1 \to -R_1, R_2 \to R_2 - R_1} \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$$

We conclude that $\left\{ \begin{bmatrix} -1 \\ 1 \end{bmatrix} \right\}$ is a basis for the eigenspace corresponding to $\lambda = 4$.

(2) We first find the characteristic polynomial:

$$p(\lambda) = \begin{vmatrix} 4-\lambda & 0 & -2 \\ 2 & 5-\lambda & 4 \\ 0 & 0 & 5-\lambda \end{vmatrix} = -(\lambda - 5)^2(\lambda - 4)$$

Thus we have $\lambda = 4, 5$ as our eigenvalues ($\lambda = 5$ is an eigenvalue with multiplicity 2). We find the corresponding eigenvectors:

($\lambda = 4$) We find a basis for the nullspace of $A - 4\lambda$:

$$\begin{bmatrix} 0 & 0 & -2 \\ 2 & 1 & 4 \\ 0 & 0 & 1 \end{bmatrix} \xrightarrow{R_1 \leftrightarrow R_2} \begin{bmatrix} 2 & 1 & 4 \\ 0 & 0 & -2 \\ 0 & 0 & 1 \end{bmatrix} \xrightarrow{R_2 \to -\frac{1}{2}R_2, R_3 \to R_3 - R_2, R_1 \to R_1 - 4R_2, R_1 \to \frac{1}{2}R_1} \begin{bmatrix} 1 & 1/2 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

We conclude that $\left\{ \begin{bmatrix} -1/2 \\ 1 \\ 0 \end{bmatrix} \right\}$ is a basis of the eigenspace corresponding to $\lambda = 4$.

($\lambda = 5$) We find a basis for the nullspace of $A - 5I$:

$$\begin{bmatrix} -1 & 0 & -2 \\ 2 & 0 & 4 \\ 0 & 0 & 0 \end{bmatrix} \xrightarrow{R_1 \to -R_1} \begin{bmatrix} 1 & 0 & 2 \\ 2 & 0 & 4 \\ 0 & 0 & 0 \end{bmatrix} \xrightarrow{R_2 \to R_2 - 2R_1} \begin{bmatrix} 1 & 0 & 2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

We conclude that $\left\{ \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -2 \\ 0 \\ 1 \end{bmatrix} \right\}$ is a basis of the eigenspace corresponding to $\lambda = 5$.

(3) It is clear that $p(\lambda) = (\lambda - 4)^2(\lambda - 2)^2$ and that the eigenvalues are $\lambda = 2, 4$, each with multiplicity 2. We now find the corresponding eigenvectors:

($\lambda = 2$) We find a basis for the nullspace of $A - 2I$:

$$\begin{bmatrix} 2 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix} \xrightarrow{R_1 \to \frac{1}{2}R_2, R_2 \to \frac{1}{2}R_2, R_4 \to R_4 - R_1} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

We conclude that $\left\{ \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} \right\}$ is a basis for the eigenspace corresponding to $\lambda = 2$.

($\lambda = 4$) We find a basis for the nullspace of $A - 4I$:

$$\begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & -2 & 0 \\ 1 & 0 & 0 & -2 \end{bmatrix} \xrightarrow{R_3 \leftrightarrow R_2, R_4 \leftrightarrow R_1, R_2 \to -\frac{1}{2}R_2} \begin{bmatrix} 1 & 0 & 0 & -2 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

We conclude that $\left\{ \begin{bmatrix} 2 \\ 0 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix} \right\}$ is a basis for the eigenspace corresponding to $\lambda = 4$.

(4) We find the characteristic polynomial:

$$p(\lambda) = \begin{vmatrix} 4 - \lambda & 3 \\ -3 & 4 - \lambda \end{vmatrix} = \lambda^2 - 8\lambda + 25 = (\lambda - 4 - 3i)(\lambda - 4 + 3i).$$

We conclude that the eigenvalues are $\lambda = 4 \pm 3i$. We first find the eigenvectors corresponding to $\lambda = 4 + 3i$:

$$\begin{bmatrix} -3i & 3 \\ -3 & -3i \end{bmatrix} \xrightarrow{R_2 \to R_2 + iR_1, R_1 \to \frac{i}{3}R_1} \begin{bmatrix} 1 & i \\ 0 & 0 \end{bmatrix}$$

We conclude that $\begin{bmatrix} -i \\ 1 \end{bmatrix}$ is an eigenvector which corresponds to $\lambda = 4 + 3i$. Taking complex conjugates, we get that $\begin{bmatrix} i \\ 1 \end{bmatrix}$ is an eigenvector which corresponds to $\lambda = 4 - 3i$.

2. *True/False. Give reasons why! Below A and B are matrices of an appropriate size.*

1. *A square matrix A is not invertible iff 0 is an eigenvalue of A.*
2. *The eigenvalues of a square matrix are the numbers on its main diagonal.*

3. *If* $\mathbf{v}_1$ *and* $\mathbf{v}_2$ *are linearly independent eigenvectors of a square matrix* A*, then they correspond to different eigenvalues.*
4. *The sum of two eigenvectors of a matrix* A *is also an eigenvector of* A*.*

1. True. It follows from the fact that the determinant is the product of eigenvalues.
2. False, see for example 3(1) and 3(4) above.
3. False, consider $A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$. A has two linearly independent eigenvectors (e.g., any basis of $\mathbb{R}^2$). However, all eigenvectors correspond to the same eigenvalue, $\lambda = 1$.
4. False. Consider $\begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}$ and eigenvectors $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$. The sum of these two eigenvectors is not an eigenvector.

3. *Let* A, B *be two* $n \times n$*-matrices such that* $AB = BA$*.*

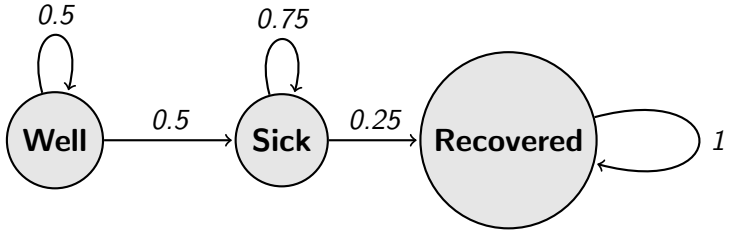
(a) Suppose $\boldsymbol{v}$ is an eigenvector of A with eigenvalue λ. Is $B\boldsymbol{v}$ an eigenvector of A? If so, what is the eigenvalue of that eigenvector?

We first must consider the case that $B\boldsymbol{v} = 0$, in which case $B\boldsymbol{v}$ cannot be an eigenvector. In the other case, consider $A(B\boldsymbol{v}) = (AB)\boldsymbol{v}$, and since $AB = BA$, this is the same as $(BA)\boldsymbol{v} = B(A\boldsymbol{v}) = B(\lambda\boldsymbol{v}) = \lambda(B\boldsymbol{v})$ (as $\boldsymbol{v}$ is an eigenvector of A with eigenvalue λ), so since $A(B\boldsymbol{v}) = \lambda B\boldsymbol{v}$, $B\boldsymbol{v}$ is an eigenvector of A with eigenvalue λ. So overall either $B\boldsymbol{v} = 0$ or $B\boldsymbol{v}$ is an eigenvector of A with eigenvalue λ_i.

(b) Suppose A has eigenvectors $\boldsymbol{v_1}, \ldots, \boldsymbol{v_n}$ with distinct eigenvalues $\lambda_1 \neq ... \neq \lambda_n$. Is each $\boldsymbol{v_i}$ also an eigenvector of B? (This question is a bit tricker. Hint: Note that each of the eigenspaces of A has dimension 1 and then use your answer to (a).).

Take the i-th eigenvalue $\boldsymbol{v_i}$ of A. Since there are n distinct eigenvalue, $\boldsymbol{v_1}, \ldots, \boldsymbol{v_n}$ form a basis of $\mathbb{R}^n$. Thus the eigenspace of A for eigenvalue λ_i is span($\boldsymbol{v_i}$). Since $\boldsymbol{v_i}$ is an eigenvector of A with eigenvalue λ_i, we get $A\boldsymbol{v_i} = \lambda_i\boldsymbol{v_i}$. Then by **a.** either $B\boldsymbol{v_i} = 0$ or $B\boldsymbol{v_i}$ is an eigenvector of A to the eigenvalue λ_i. If $B\boldsymbol{v_i} = 0$, then $\boldsymbol{v_i}$ is eigenvector of B with eigenvalue 0. So it is left to consider the case that $B\boldsymbol{v_i}$ is an eigenvector of A to the eigenvalue λ_i. Then $B\boldsymbol{v_i}$ is in the eigenspace of A of the eigenvalue λ_i. Then $B\boldsymbol{v_i}$ is in the span of $\boldsymbol{v_i}$. Therefore $B\boldsymbol{v_i}$ is a multiple of $\boldsymbol{v}_i$. Hence $\boldsymbol{v_i}$ is an eigenvector of B.

4. *Suppose there is an epidemic with the following transition graph:*



Set up a 3×3 *transition matrix* A *and find the steady state* $\boldsymbol{v}_\infty$*.*

Hence, the transition matrix is

$$A = \begin{bmatrix} .5 & 0 & 0 \\ .5 & .75 & 0 \\ 0 & .25 & 1 \end{bmatrix}.$$

Let

$$u_t = \begin{bmatrix} u_{t,W} & u_{t,S} & u_{t,R} \end{bmatrix},$$

where $u_{t,W}$ is the percentage of people that are well, $u_{t,S}$ is the percentage of people that are sick and $u_{t,R}$ the percentage of people that recovered, at time t. Note that

$$u_{t+1} = Au_t.$$

The long term equlibrium is a state u_∞ such that

$$Au_\infty = u_\infty.$$

For that we need to find an eigenvector of A to the eigenvalue 1:

$$\left[\begin{array}{ccc|c} -.5 & 0 & 0 & 0 \\ .5 & -.25 & 0 & 0 \\ 0 & .25 & 0 & 0 \end{array}\right] \to \left[\begin{array}{ccc|c} -.5 & 0 & 0 & 0 \\ 0 & -.25 & 0 & 0 \\ 0 & .25 & 0 & 0 \end{array}\right] \to \left[\begin{array}{ccc|c} -.5 & 0 & 0 & 0 \\ 0 & -.25 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right] \to \left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

Hence, the eigenspace of $\lambda = 1$ is spanned by $\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$. Since this vector is already scaled so that its entries add up to 1, we conclude

$$u_\infty = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}.$$

This means all people will recover from the epidemic.